

Exposé: Toward a Fully Automated Migration Framework from Windows 11 to Linux Mint

1. Introduction

The migration from Microsoft Windows 11 to Linux Mint can be challenging, especially for users without strong technical knowledge. Tasks such as identifying alternative applications, selecting files for migration, and ensuring a safe transfer process often require manual effort and can lead to errors.

A semi-automated migration system has already been developed to guide users through this process. It provides a structured, step-by-step workflow including system scanning, application mapping, backup, restore, and validation. While this reduces complexity, the system still depends on user interaction at several stages.

This Praktikum project aims to extend the existing system toward a more automated workflow, while maintaining clarity, user control, and data privacy.

2. Problem Statement

The current system is functional and user-friendly but not fully automated. The main limitations are:

- Application mappings are primarily based on static configuration sources (e.g., CSV mappings), limiting adaptability to diverse user requirements
- Although customization options exist, the system still requires explicit user involvement, and the balance between automation and user control is not yet fully optimized
- The validation component provides a status-based report, but lacks detailed diagnostic information for deeper analysis
- Some parts of the system (e.g., CLI workflow) are not yet fully integrated with the centralized automation and policy logic

In addition, the system must ensure strict privacy constraints:

- User data must not be transmitted externally
- External services, if used, must be limited to non-sensitive metadata queries

3. Objectives

The goal of this project is to extend the existing migration system toward a more automated and adaptive workflow. The main objectives are:

- Improve application mapping by enhancing config-based mappings with more dynamic and adaptable strategies
- Strengthen recommendation mechanisms for:
 - Application selection
 - File prioritization
- Integrate external metadata sources such as Repology to improve application availability insights
- Enhance the validation component to provide more detailed and informative reports
- Improve integration between system components, particularly ensuring consistency between Qt-based workflows and CLI execution
- Increase the level of automation while preserving user control through guided, balanced, and expert modes
- Ensure privacy-preserving system behavior, where:
 - No user data is transmitted externally
 - External API usage is limited to non-sensitive metadata

4. System Overview

The system consists of the following main components:

- Inventory module: collects information about installed applications and file structures
- Recommendation module: suggests application mappings and important files
 - Uses local configuration and optional AI-based ranking
 - External API usage (e.g., Repology) is limited to software metadata
- Workflow controller: manages execution flow and user interaction modes
- Migration module: handles backup and restore operations
- Validation module: generates reports showing the success or failure of migration steps

The system supports both a graphical interface (Qt-based wizard) and a command-line interface.

5. Methodology

The project will be carried out in the following steps:

1. System refinement
 - Refine system architecture and improve module interaction
2. Recommendation enhancement
 - Improve mapping and file selection strategies
 - Ensure correct handling of AI configuration and privacy
3. Workflow integration

- Connect all components into a more automated pipeline
 - Improve consistency between UI and CLI workflows
4. Testing and validation
- Perform end-to-end migration tests
 - Verify correctness of backup, restore, and recommendations
5. Documentation
- Provide technical and user-oriented documentation

6. Work Breakdown Structure

The project is divided into several work packages covering system refinement, recommendation enhancement, workflow integration, validation improvements, testing, and documentation.

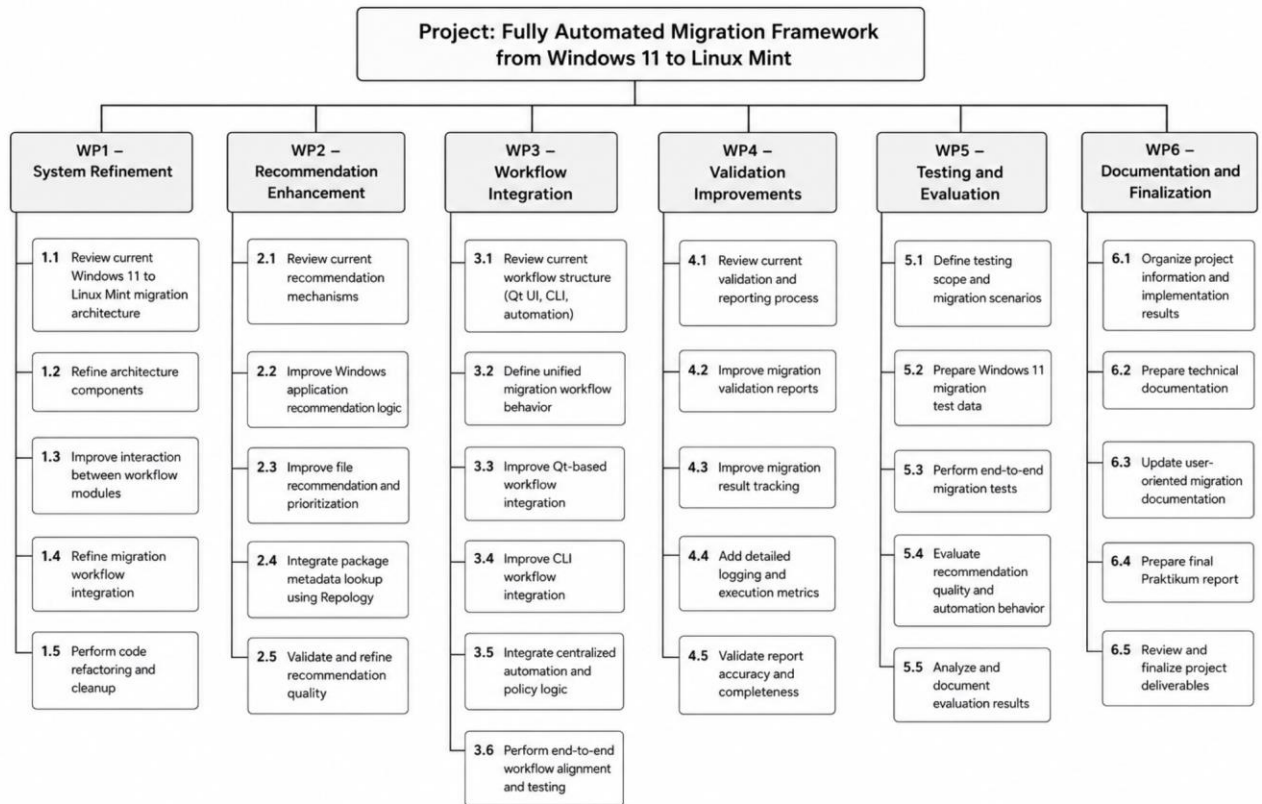


Figure 1: Work Breakdown Structure of the Windows 11 to Linux Mint Migration Assistant Project

7. Project Timeline

The project timeline is organized across eight weeks. The implementation progresses from system refinement and recommendation enhancement toward workflow integration, validation, testing, and final documentation.

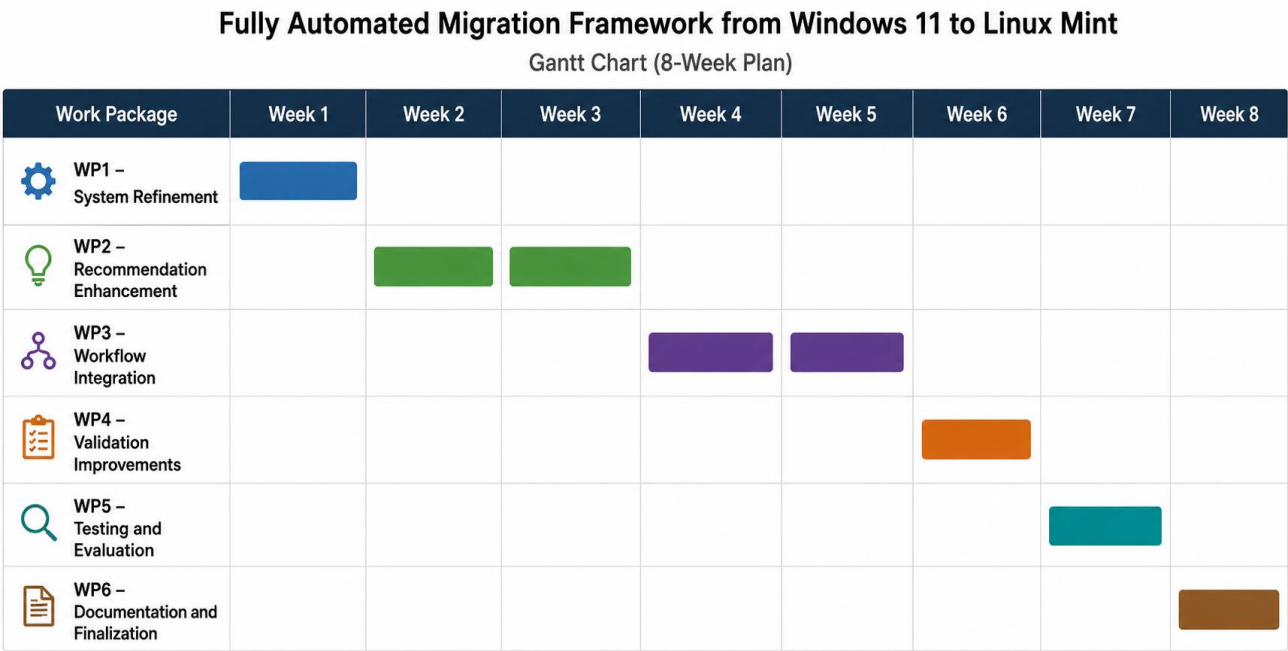


Figure 2: Gantt Chart of the Planned Project Timeline

8. Evaluation Plan

The system will be evaluated based on:

- Functionality: correctness of migration steps
- Recommendation quality: usefulness of suggested applications and files
- Automation level: reduction of required user interaction
- Performance: execution time and resource usage
- Reliability: stability and correctness of validation reports
- Privacy: ensuring no user data is transmitted externally

9. Current Progress

The system has already reached an advanced development stage, forming a strong foundation for further automation. The following components have been implemented:

- A Qt-based wizard interface and command-line interface for user interaction
- A modular architecture separating workflow control, recommendation logic, and

execution components

- Application mapping based on configuration files with optional metadata enrichment via Repology
- File recommendation functionality using local scoring and optional AI-based ranking (policy-controlled)
- A mode-driven workflow supporting guided, balanced, and expert user interaction
- Backup and restore mechanisms for file migration
- A validation component that generates status-based migration reports

The current implementation provides a functional semi-automated workflow. The remaining work focuses on improving automation, integration, and recommendation quality.

10. Expected Outcome

The project will result in an improved migration system that:

- Provides a more automated workflow
- Offers better recommendations for applications and files
- Maintains user control and transparency
- Ensures data privacy and safe migration

11. Conclusion

This project extends an existing semi-automated migration system toward a more automated and adaptive solution. The focus is on improving workflow integration, recommendation quality, and validation capabilities while ensuring reliability and privacy. The resulting system will provide practical support for users migrating from Windows 11 to Linux Mint.